



Book I. Prop. IV. Theor.

If two triangles have two sides of the one respectively equal to two sides of the other, (— to — and — to —) and the angles (◡ and ◡) contained by those equal sides also equal; then their bases or their sides (— and —) are also equal: and the remaining angles opposite to equal sides are respectively equal (◡ = ◡ and ◡ = ◡): and the triangles are equal in every respect.

Let two triangles be conceived, to be so placed, that the vertex of one of the equal angles (◡ or ◡) shall fall upon that of the other, and — to coincide with —, then will — coincide with — if applied: consequently — will coincide with —, or two straight lines will enclose a space, which is impossible (ax. X), therefore — = —, ◡ = ◡ and ◡ = ◡,

and as the triangles ◡ and ◡ coincide, when applied, they are equal in every respect.

Q. E. D.